

Dynamic Operation of Buck-Boost DC-DC Converters over Wide Operating Ranges with Switching Based Model Predictive Control (MPC)

Min Jeong and Jürgen Biela

Laboratory for High Power Electronic Systems (HPE), ETH Zürich
jeong@hpe.ee.ethz.com

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Abstract

In this paper, a novel model predictive control (MPC) scheme is proposed for buck-boost converters for achieving a high control bandwidth over wide operating ranges. There, the duty cycles of the four possible switching states are considered as control inputs, such that an exact state propagation can be accurately predicted in a large-signal linear model. Unlike conventional methods, which suffer from poor dynamic control performance when the input voltage is close to the output voltage, the proposed method achieves good dynamic and steady-state control performance independent of the operating point by actively utilizing all four switches during transients and switching only two switches during the steady-state. The resulting quadratic programming (QP) problem enables a real-time implementation of the proposed MPC method at a fast sampling rate.

1 Introduction

In DC systems, such as battery energy storage [1] or photovoltaic (PV) systems [2], a flexible operation of DC-DC converter is required, since the voltage levels at one or both side of the converter can change in a wide range and may vary abruptly [3], [4]. Non-inverting buck-boost converters as for example shown in Fig. 1(a) are a widely used topology for such a power conversion due to its high power density and efficiency. Recently, this topology is further used as an Y-inverter replacing a common DC/DC boost converter cascaded with a voltage source inverter [5].

Designing a controller for such a converter that can operate in a wide range is not trivial with conventional design methods. The most common and popular controller design method is based on a small-signal linearized transfer function, since the conventional control theories can be easily applied [6], [7]. However, the frequency characteristics of the small-signal plant model depend on the operating point and the plant changes a lot, particularly when both input and output voltage vary in a wide range [8], [9]. Furthermore, buck-boost converters have additional challenges, including a right-half plane (RHP) zero due to non-minimum phase behavior during the boost mode and control-mode transition when the input and output voltage are close to each other [10], [11].

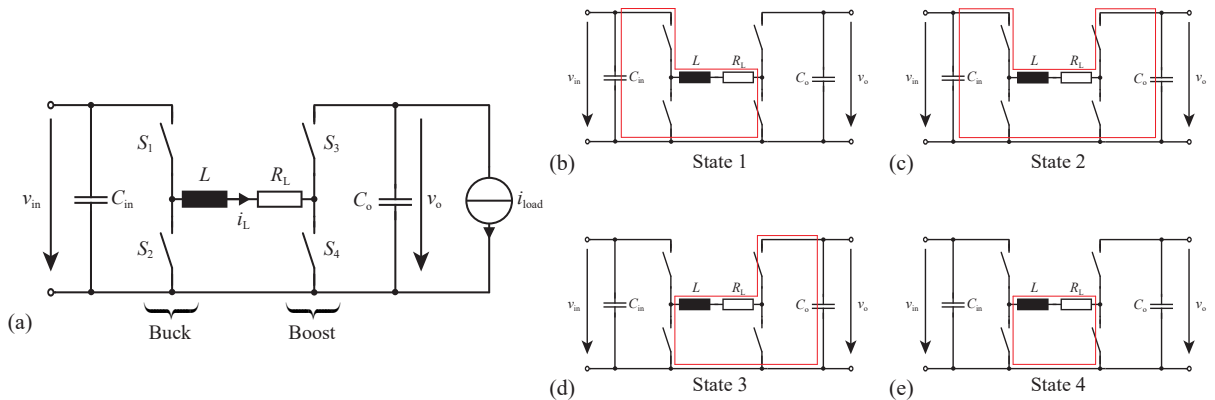


Fig. 1: (a) Considered non-inverting buck-boost converter. (b)-(e) Four switching states of the buck-boost converter.

In more and more applications, a faster dynamic response of the DC-DC converter is required. With a fast controller, the need for large electrolytic DC link capacitors, which are bulky, expensive, and unreliable, can be avoided for achieving a regulated DC-link voltage during transient situations [12]. Moreover, in PV systems, it is required to cope with fast maximum power point tracking and mismatches between panels, which could be caused by for example partial shading [2], [8]. However, the requirement of a good dynamic performance further aggravates the controller design. A few attempts to design fast controllers have been made based on MPC methods [13], [14]. However, the averaged large-signal model of the converter shows nonlinearities, making the adoption of predictive control difficult. Therefore, the model is typically linearized by approximating the model around the operating point in [14] or by fixing the duty cycles of the boost stage as a piecewise-constant function as presented in [13]. In all cases, the operating ranges and the performance are limited.

In this paper, a novel MPC controller is proposed to overcome these limitations and to offer a dynamic and flexible operation of the buck-boost converter in all operating ranges. In contrast to conventional control methods, duty cycles of each switching state are explicitly computed, so that a MPC law can be formulated without linearizations at certain operating points. The achieved MPC law results in a quadratic programming (QP) problem.

The paper is organized as follows: section 2 reviews conventional models and provides a switching based modeling of the buck-boost converter, and then section 3 introduces a control structure with an MPC formulation for the inner-current control based on the derived model. Simulation results are presented in section 4 to evaluate the performance of the proposed MPC method compared to conventional methods. Conclusions are given in section 5.

2 Buck-boost DC-DC Converter Operation and Model

A typical non-inverting buck-boost converter is shown in Fig. 1(a), which basically consists of a buck and a boost converter in series. Depending on the application, either IGBTs or MOSFETs can be used as switches and two or four switches can be utilized for uni or bi-directional operation. The buck-boost converter is used in many applications due to its versatility and many advantages, and large amount of research efforts have been made for modeling the converter and improving the control performance. In this section, common models of the buck-boost converter are reviewed in section 2.1 and then a novel hybrid model with a constraint is proposed in section 2.2.

2.1 Limitations of conventional models

The operation of the considered buck-boost converter can be split into the four different modes as shown in Fig. 1(b)-(e). The large-signal converter modelling is typically performed by averaging the four modes over a switching interval as

$$\frac{di_L(t)}{dt} = \frac{d_{\text{buck}}(t)v_{\text{in}}(t) - R_L i_L(t) - d_{\text{boost}}(t)v_o(t)}{L} \quad \text{and} \quad \frac{dv_o(t)}{dt} = \frac{d_{\text{boost}}(t)i_L(t) - i_{\text{load}}(t)}{C_o} \quad (1)$$

where the duty cycle of the buck and boost stage (d_{buck} , d_{boost}) are chosen as control inputs. Despite the simplified converter model in a continuous time-domain by neglecting switching actions, the control of the resulting averaged

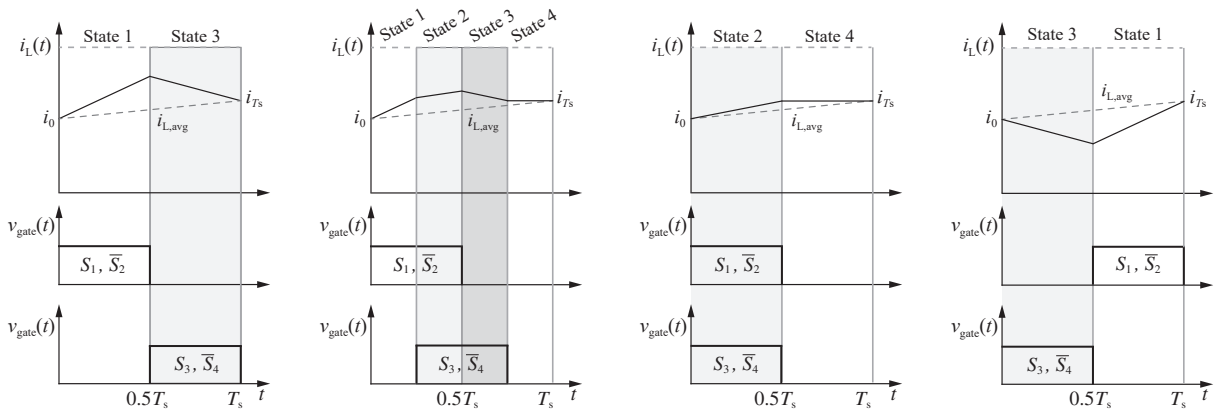


Fig. 2: Different possible inductor current waveforms when same duty cycles ($d_{\text{buck}} = 0.5$, $d_{\text{boost}} = 0.5$) are applied with different modulation sequences. Grey-colored areas indicate states that transfer energy to the output capacitor. It can be observed that different amounts of energy flow to the output capacitor depending on the modulation sequences. The effect becomes more pronounced when the inductance becomes smaller.

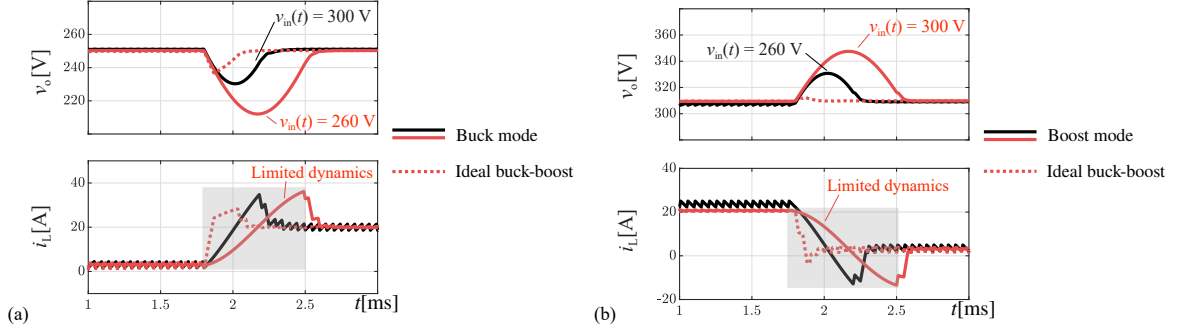


Fig. 3: Limitations of multi-mode operation when the input voltage is close to the output voltage. (a) In buck mode ($v_o = 250$ V), a limited dynamic is achieved for a load step-up. (b) In boost mode ($v_o = 310$ V), a limited dynamic is achieved for a load step-down. In both cases, improved dynamics can be achieved with smaller voltage fluctuations when the buck-boost mode (use of all four switches) is utilized.

model is still not trivial, since the model is nonlinear due to multiplications of the states and the control input (boost duty cycle). Moreover, the averaging covers up the exact switching behavior of the converter with diverse possible modulation sequences as shown in Fig. 2. Different modulation sequences can be attained even with the same control inputs (d_{buck} , d_{boost}) based on considered modulation schemes. Proper modulations are necessary not only for achieving the required dynamic control performance but also for reducing the ripple and/or the average inductor current during steady-state operations [7].

To overcome the difficulties due to the nonlinearity, commonly a multi-mode approach is used, which operates the buck-boost converter either in buck mode or in boost mode depending on the operating voltage. Since only two switches are switched (S_1 and S_2 in the buck mode and S_3 and S_4 in the boost mode), this method typically results in a higher efficiency and a simpler design process of a controller based on well-formulated controller design rules for the buck/boost converter. However, the multi-mode approach still has a few disadvantages. First, the dynamic performance of the controller is limited when the input voltage is close to the output voltage, since only two switches are actively utilized [9]. The dynamic performance can be dramatically improved as shown in Fig. 3 when all four switches are used for switching. Second, a transition mode between buck and boost mode is required in practical implementation [15], [16]. This is because there are limitations on the duty cycles due to the dead time between switches and propagation delays. Many methods are proposed to address this problem, but this complicates the controller design process due to additional design parameters (transition range and number of sub-zones in the range) and still results in a sub-optimal behavior.

2.2 Hybrid model

Instead of averaging the operations over a switching interval and choosing the buck and the boost duty cycles as control inputs, the duty cycles of each state are chosen as control inputs in this paper. In this way, the nonlinearity of the averaged model can be avoided and the exact state propagation can be predicted by a linear model over one switching interval. The four states can be represented as a hybrid system with a linear state-space model as follows:

$$\frac{d\mathbf{x}(t)}{dt} = \mathbf{A}_i \cdot \mathbf{x}(t) + \mathbf{F}_i \cdot \mathbf{w}(t) \quad (2)$$

where $\mathbf{x}(t) = [v_o(t) \quad i_L(t)]^T$, $\mathbf{w}(t) = [v_{in}(t) \quad i_{load}(t)]^T$, and the matrices are given by

$$\begin{cases} \mathbf{A}_1 = \begin{bmatrix} 0 & 0 \\ 0 & -\frac{R_L}{L} \end{bmatrix} & \mathbf{F}_1 = \begin{bmatrix} 0 & -\frac{1}{C_o} \\ \frac{1}{L} & 0 \end{bmatrix} & \text{Energy transfer between input source and inductor (State 1)} \\ \mathbf{A}_2 = \begin{bmatrix} 0 & \frac{1}{C_o} \\ -\frac{1}{L} & -\frac{R_L}{L} \end{bmatrix} & \mathbf{F}_2 = \begin{bmatrix} 0 & -\frac{1}{C_o} \\ \frac{1}{L} & 0 \end{bmatrix} & \text{Direct energy transfer between input and output source (State 2)} \\ \mathbf{A}_3 = \begin{bmatrix} 0 & \frac{1}{C_o} \\ -\frac{1}{L} & -\frac{R_L}{L} \end{bmatrix} & \mathbf{F}_3 = \begin{bmatrix} 0 & -\frac{1}{C_o} \\ 0 & 0 \end{bmatrix} & \text{Energy transfer between inductor and output source (State 3)} \\ \mathbf{A}_4 = \begin{bmatrix} 0 & 0 \\ 0 & -\frac{R_L}{L} \end{bmatrix} & \mathbf{F}_4 = \begin{bmatrix} 0 & -\frac{1}{C_o} \\ 0 & 0 \end{bmatrix} & \text{Hold state (State 4)} \end{cases}$$

Note that $\mathbf{w}(t)$ is not a control input and the value of $\mathbf{w}(t)$ is already known through measurements or estimators. All bold and capital letters represent matrices, and all bold and lower-case letters represent vectors. Since the switch should be turned on/off only once during a switching interval, there are only two possible sequences of states, which are the sequence (1-2-3-4) and the reverse (4-3-2-1). However, this has a negligible impact on the control performance.

By taking the forward-Euler method with a duty cycle D_i for each mode, (2) can be discretized as

$$\mathbf{x}[k+D_1] = \mathbf{x}[k] + T_s \cdot D_1 \cdot (\mathbf{A}_1 \cdot \mathbf{x}[k] + \mathbf{F}_1 \cdot \mathbf{w}[k]) \quad (3a)$$

$$\mathbf{x}[k+D_1+D_2] \approx \mathbf{x}[k+D_1] + T_s \cdot D_2 \cdot (\mathbf{A}_2 \cdot \mathbf{x}[k] + \mathbf{F}_2 \cdot \mathbf{w}[k]) \quad (3b)$$

$$\mathbf{x}[k+D_1+D_2+D_3] \approx \mathbf{x}[k+D_1+D_2] + T_s \cdot D_3 \cdot (\mathbf{A}_3 \cdot \mathbf{x}[k] + \mathbf{F}_3 \cdot \mathbf{w}[k]) \quad (3c)$$

$$\mathbf{x}[k+D_1+D_2+D_3+D_4] = \mathbf{x}[k+1] \approx \mathbf{x}[k+D_1+D_2+D_3] + T_s \cdot D_4 \cdot (\mathbf{A}_4 \cdot \mathbf{x}[k] + \mathbf{F}_4 \cdot \mathbf{w}[k]) \quad (3d)$$

Since T_s is small, the second order terms are neglected and $\mathbf{w}[k]$ is assumed to be constant during a switching interval. With this model and the condition, $D_1 + D_2 + D_3 + D_4 = 1$, the buck-boost converter can be modelled for all operating conditions.

Generally, DC-DC converters are usually designed to have enough DC link/output capacitance, such that voltage fluctuations are limited and the current prediction model is valid. However, depending on the inductance value, the voltage prediction model may not be accurate enough, since the first-order approximation (forward-Euler method) does not take into account the variation during the states. Therefore, unless a long-prediction MPC is utilized, this results into a degraded control performance. Therefore, an MPC is proposed only for the inner current controller in this paper, such that it can accurately predict exact state propagation for all converter parameters at all operating points. Details about the overall control structure is described in the next section.

3 Control Design

Based on the derived buck-boost converter model, a formulation for MPC is presented in the following. Ideally, a long prediction horizon is required to handle a non-minimum phase behavior during the boost mode because the energy transfer is conducted indirectly by storing the energy first in the inductor. However, a long prediction horizon dramatically increases the computational effort, which often limits the practical usage of MPC methods. Also, long-horizon MPC results in a non-convex optimization problem even with the hybrid model, since the approximation in (3) is no longer valid for a long-horizon prediction and the state propagation equality constraints should contain multiplications of the input (D) and the state ($\mathbf{x}$). Therefore, this results in a complex quadratic optimization with quadratic equality constraints. Instead, MPC with a one-step prediction horizon is proposed as an inner-current controller in this paper, and a PI-controller is utilized as an outer-voltage controller as shown in Fig. 4. Nevertheless, the overall control performance improves and performs similar as an ideal MPC with a long-prediction horizon, since the bandwidth of the MPC current controller is higher at all operating points even with constraints taken into account.

In power electronic applications, a finite control set (FCS) MPC is widely introduced, because it is simple, intuitive, and powerful. However, FCS-MPC complicates the converter component design process since the switching frequency is varying and also gets influenced by control parameters. In contrast, a continuous control set (CCS)

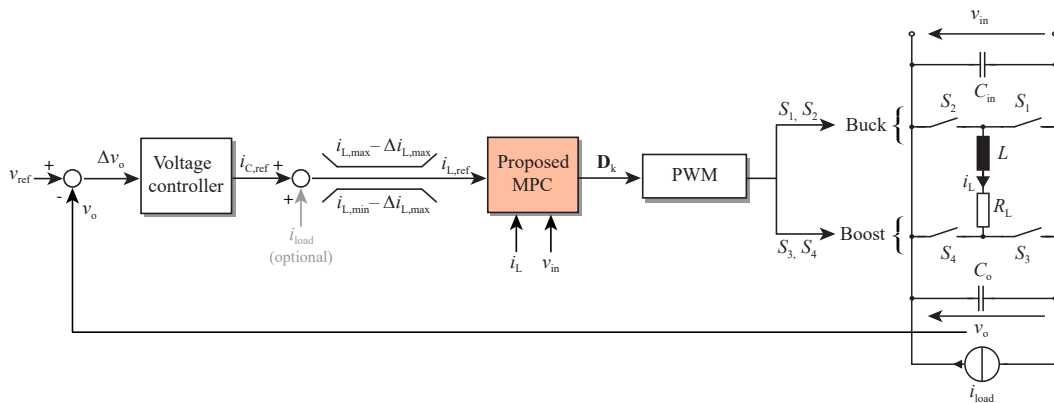


Fig. 4: Control system diagram with the proposed MPC for the buck-boost converter. Unlike the multi-mode approach, the proposed MPC has a single unified current controller that operate at all points.

MPC, as the proposed one, has a fixed switching frequency and operates in a similar framework as conventional control methods. Therefore, controller parameters do not influence the switching frequency, and the component designs and the controller parameter tuning can be done separately based on the fixed switching frequency. There, all well-known optimal converter design methods can be still applied, and MPC can simply bring advanced control performance to the converter system.

3.1 Proposed Current Controller of Switching Based MPC (SMPC)

With the discretized model in (3), the progression of the states over one switching interval can be described by a linear model as

$$\mathbf{x}_f[k] = \begin{bmatrix} \mathbf{x}[k+D_1] \\ \mathbf{x}[k+D_1+D_2] \\ \mathbf{x}[k+D_1+D_2+D_3] \\ \mathbf{x}[k+D_1+D_2+D_3+D_4] \end{bmatrix} = \underbrace{\begin{bmatrix} \mathbf{x}[k] \\ \mathbf{x}[k] \\ \mathbf{x}[k] \\ \mathbf{x}[k] \end{bmatrix}}_{\mathbf{x}_0[k]} + \underbrace{\begin{bmatrix} \mathbf{G}_1 & \mathbf{0}_{2 \times 1} & \mathbf{0}_{2 \times 1} & \mathbf{0}_{2 \times 1} \\ \mathbf{G}_1 & \mathbf{G}_2 & \mathbf{0}_{2 \times 1} & \mathbf{0}_{2 \times 1} \\ \mathbf{G}_1 & \mathbf{G}_2 & \mathbf{G}_3 & \mathbf{0}_{2 \times 1} \\ \mathbf{G}_1 & \mathbf{G}_2 & \mathbf{G}_3 & \mathbf{G}_4 \end{bmatrix}}_{\mathbf{G}[k]} \cdot \underbrace{\begin{bmatrix} D_1 \\ D_2 \\ D_3 \\ D_4 \end{bmatrix}}_{\mathbf{D}[k]} \quad (4)$$

where $\mathbf{x}_f[k]$ is the future state vector, $\mathbf{x}_0[k]$ is the current state vector, $\mathbf{G}_i[k] = T_s \cdot (\mathbf{A}_i \cdot \mathbf{x}[k] + \mathbf{F}_i \cdot \mathbf{w}[k])$ is the discrete relation matrix, $\mathbf{0}_{m \times n}$ represents a zero matrix, and $\mathbf{D}[k]$ is the control input vector. Note that $\mathbf{x}_0[k]$ and $\mathbf{G}[k]$ are varying constants for each switching interval.

The MPC law for directly obtaining the optimal duty cycles $\mathbf{D}[k]$ can be written as

$$\min_{\mathbf{D}[k]} \left\| i_L[k+D_1+D_2] - i_{L,\text{ref}} \right\|_{\mathbf{Q}}^2 + \left\| i_L[k+D_1+D_2+D_3] - i_{L,\text{ref}} \right\|_{\mathbf{Q}}^2 + \left\| D_4 \right\|_{\mathbf{R}}^2 \quad (5a)$$

$$s.t. \quad \mathbf{x}_f[k] = \mathbf{x}_0[k] + \mathbf{G}[k] \cdot \mathbf{D}[k] \quad (\text{Equality constraint}) \quad (5b)$$

$$\mathbf{1} = \mathbf{1}_{1 \times 4} \cdot \mathbf{D}[k] \quad (\text{Duty cycle summation constraint}) \quad (5c)$$

$$\mathbf{0}_{4 \times 1} \leq \mathbf{I}_4 \cdot \mathbf{D}[k] \quad (\text{Nonnegative duty cycle constraint}) \quad (5d)$$

$$i_L[k+D_1] \leq i_{L,\text{max}} \quad (\text{Max. current constraint on state 1 from Fig. 1}) \quad (5e)$$

$$i_{L,\text{min}} \leq i_L[k+D_1+D_2] \leq i_{L,\text{max}} \quad (\text{Max. and min. current constraint on state 2 from Fig. 1}) \quad (5f)$$

$$i_{L,\text{min}} \leq i_L[k+D_1+D_2+D_3] \quad (\text{Min. current constraint on state 3 from Fig. 1}) \quad (5g)$$

where $\mathbf{Q}, \mathbf{R} \geq 0$ are weighting matrices, $\|\mathbf{z}\|_{\mathbf{Q}}^2$ denotes a 2-norm with the weighting matrix $\mathbf{Q}$, and $\mathbf{1}_{m \times n}$ and $\mathbf{I}_m$ represent a matrix of ones and identity matrix respectively. Note that both the maximum and the minimum current constraints are applied at state 2 since this state can increase or decrease the inductor current depending on operating points.

With the proposed formulation, one MPC can handle both buck and boost mode and there is no need to consider mode-transition. Therefore, multiple formulations which complicate the controller implementation, i.e. additional controller design parameters for deciding what is the best transition range, can be avoided. Furthermore, other practical limitations can be easily resolved with the proposed method by putting additional constraints on duty cycles, e.g. minimum duty cycle length due to dead-time and/or gate driver delays. Finally, the resulting optimization problem is a quadratic programming (QP), such that a real-time MPC can be implemented at a high sampling rate (in the order of 20kHz or more) [17].

3.2 Outer Voltage Controller

The outer voltage controller can be designed similar to a normal multi-mode cascaded PI-controller for the buck-boost converter. In [5] a guideline for the controller parameter tuning is suggested, which is summarized in Table

Table I: Controller parameter tuning guidelines from [5].

Controller type	Voltage	Current
Plant transfer function	$G_v = \frac{1}{sC}$	$G_i = \frac{1}{sL}$
Controller transfer function	$C_v = K_v \frac{1+sT_v}{sT_v}$	$C_i = K_i \frac{1+sT_i}{sT_i}$
Crossover frequency	$f_v = \frac{f_i}{10} = \frac{f_s}{100}$	$f_i = \frac{f_s}{10}$
Proportional gain	$K_v = 2\pi f_v C$	$K_i = 2\pi f_i L$

I. The main difference with the MPC is that the current crossover frequency can be directly computed more accurately based on the system parameters. The current bandwidth is guaranteed even when the input voltage is close to output voltage, and this enables a higher bandwidth of the overall controller at all operating points. The current and voltage crossover frequency can be computed as:

$$f_i = \frac{\min(v_{in}, v_o)}{L \cdot (i_L^{\max} - \Delta i_L^{\max})} \quad f_v = \frac{f_i}{5}$$

With the derived equations and table, all controller parameters in the following section can be computed.

4 Simulation Results

In this section, the performance of the proposed switching based MPC (SMPC) method is evaluated with simulations and compared to two existing methods: the multi-mode approach with voltage and current dual-loop control (DLC) method [5] and the linear MPC (LMPC) method [13]. The buck-boost converter parameters for the simula-

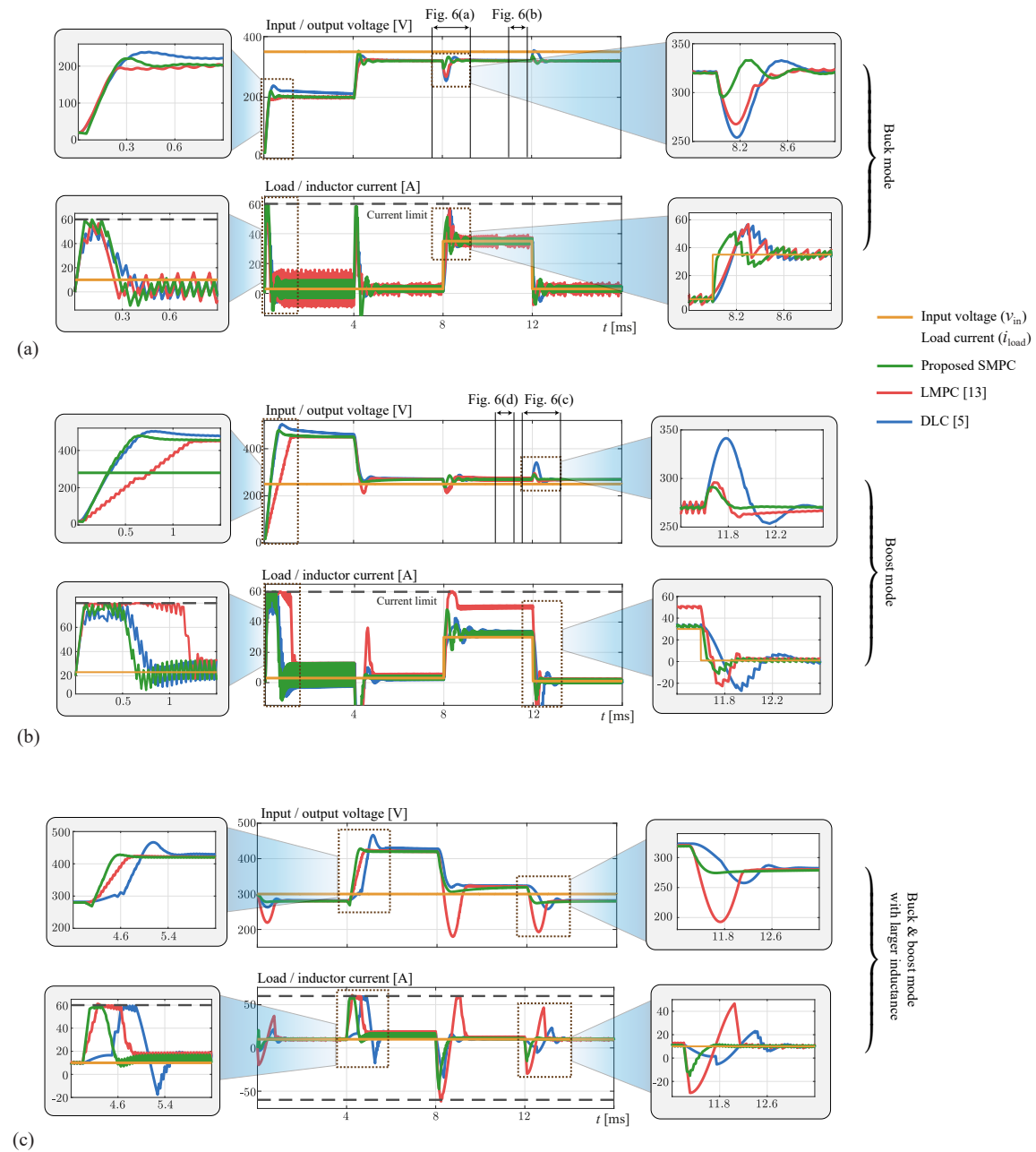


Fig. 5: Simulation results for comparing the three control methods for (a) Buck mode, (b) Boost mode, and (c) Buck and boost mode with a bigger inductance ($2L = 732\mu\text{H}$) compared to the parameter in [18].

Table II: Buck-boost converter parameters.

Symbol		Value	Symbol		Value
P	Rated power	10kW	f_{sw}	Switching frequency	20kHz
v_{in}	Input voltage	250-350 V	v_o	Output voltage	190V-450 V
L	Inductance	366 μ H	R_L	Lumped series resistance	0.02 Ω^*
C_o	Output capacitance	50 μ F	$i_L^{\max}$	Inductor saturation current	60 A *

* These parameters are not specified in [18] and are assumed based on other parameters.

tions are based on the parameters in [18] and are given in Table II. For comparisons of the dynamic performance of the controllers, the transient behavior, a start-up and a load transient, are evaluated. Most controller parameters are designed referring to the guidelines given in section 3.2, but a simple P-controller is used instead of a PI-controller for the inner current control of the DLC method. The parameters used in the simulations are given in Table III.

The simulation results of the buck mode are shown in Fig. 5(a). First, a start-up behavior is compared, followed by an output voltage step at $t = 4$ ms. The load changes at $t = 8$ ms and returns back at $t = 12$ ms. Compared to the DLC method, the SMPC shows a faster transient behavior for tracking the reference with a smaller overshoot and reduced voltage variations for load changes. The LMPC shows very good control performance for the start-up and reference step because the applied approximation of the piecewise-constant function is very effective for the buck operation. However, the LMPC still performs less well similar as the DLC method at load changes since only two switches are allowed to operate. In Fig. 6(a), the detailed transient behaviors are illustrated with switching

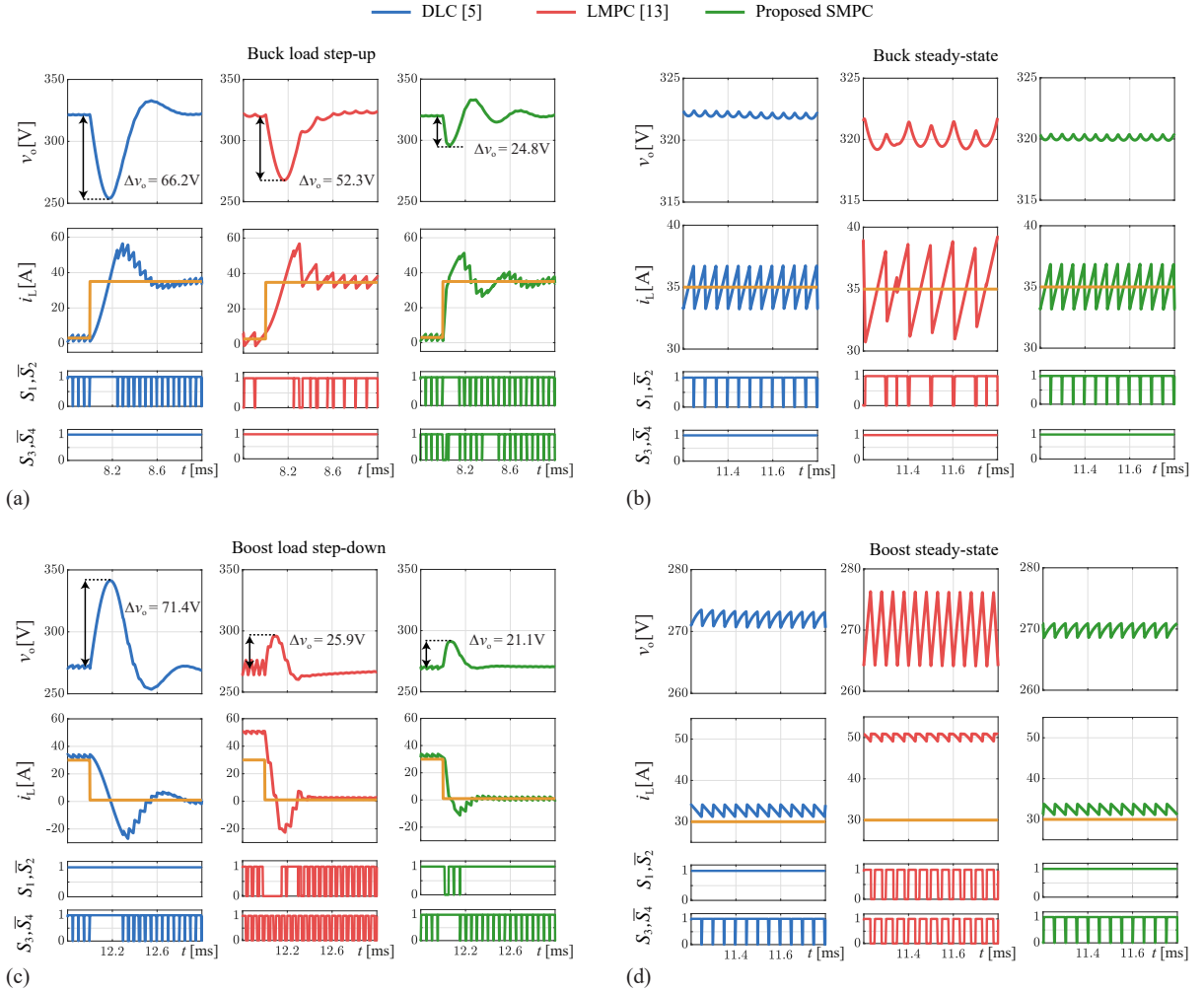


Fig. 6: Magnified simulation results of Fig. 5 for comparing the three control methods with detailed switching commands. (a) Transient behaviors (b) Steady-state behaviors for the buck mode, and (c) Transient behaviors (d) Steady-state behaviors for the boost mode.

Table III: Controller parameters.

Symbol	Parameter	Default parameters		Large inductance in Fig. 5(c)	
		SMPC	DLC	SMPC	DLC
$\mathbf{Q}$	Penalty on state error	1	-	1	-
$\mathbf{R}$	Penalty on control input	1×10^4	-	1×10^4	-
$\mathbf{K}_i$	Current proportional gain	-	6.90	-	13.80
$\mathbf{K}_i/T_i$	Current integral gain	-	0	-	0
$\mathbf{K}_v$	Voltage proportional gain	0.54	0.63	0.27	0.63
$\mathbf{K}_v/T_v$	Voltage integral gain	1411	126	353	126
f_s	Sampling frequency	20 kHz*			

* Sampling frequency is equal to the switching frequency. In practical implementation, the unit-delay method as shown in [19] is typically applied to compensate time-delays of measurements, computations, and communications.

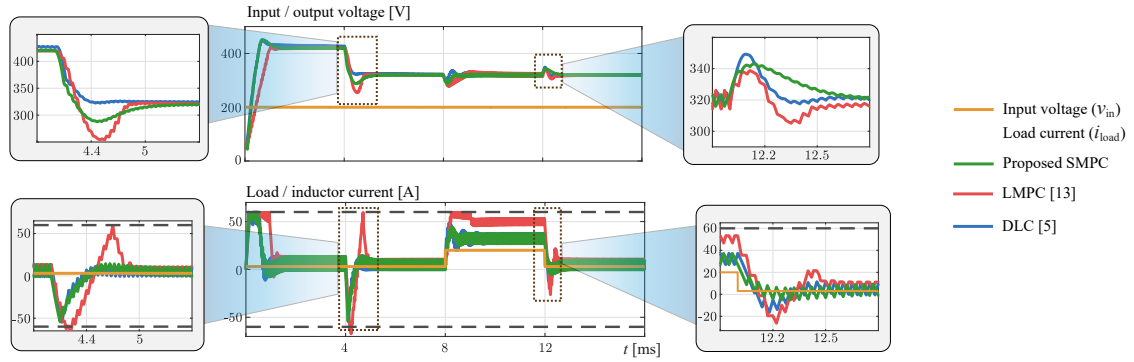


Fig. 7: Simulation results for comparing the three control methods for a normal boost mode when the input and output voltage are not close to each other.

commands. The output voltage variation of the SMPC (24.8 V) is smaller by more than 50% compared to the DLC (66.2 V) and the LMPC (52.3 V). Furthermore, in Fig. 6(b), detailed steady-state behaviors are illustrated, and it can be observed that the LMPC performs less well with larger output voltage ripples and inductor current ripples because of the averaged model. Unlike that, the SMPC reaches a steady-state faster with less ripples and performs like a normal buck mode with a good power efficiency.

The simulation results of the boost mode are shown in Fig. 5(b) with a output voltage step and load changes. For the DLC and SMPC methods, similar results as the buck mode are observed, i.e. the SMPC shows a faster transient behavior for tracking a reference with a smaller overshoot and reduced voltage variations for load changes. On the other hand, the performance of the LMPC is limited even during a start-up and a voltage step, since the applied approximation fixes the duty cycle of the boost stage and hinders a dynamic behavior. Furthermore, this results in a sub-optimal steady-state behavior as depicted in Fig. 6(d). The DLC and SMPC methods achieve a small averaged inductor current (32.5 A), but the LMPC has a larger averaged inductor current (50.1 A) and suffers from higher power losses and bigger output voltage ripples. In terms of the output voltage fluctuation during a load step-down, the SMPC (21.1 V) still shows the best performance and achieves 70% less fluctuations compared to the DLC method (71.4 V) as shown in Fig. 6(c).

The simulation result with a higher inductance value ($2L = 732 \mu\text{H}$) is shown in Fig. 5(c) for investigating the effect of parameters on the dynamic control performance. The output voltage reference changes such that the converter operates both in the buck and the boost region. As the inductance becomes larger, degraded control performance of the DLC method becomes more obvious when the input and the output voltage are close. This results in that the settling time of the SMPC method is smaller by a factor of four or more compared to that of the DLC method.

For a complete overview on control performance, simulation results for a normal operation, when the input and output voltage are not close, are given in Fig. 7. Unlike the results in Fig. 5(b), the DLC method can achieve a sufficient dynamic performance since there is enough voltage difference between the input and the output voltage. It can be noted that the SMPC performs similar to the DLC method in such an operating range, when using four switches does not give additional benefits. Therefore, the SMPC can be regarded as an advanced version of the DLC method that exploits the full dynamic potential of the converter with all four switches if necessary for

achieving the fastest possible responses.

5 Conclusion

This paper presents a novel MPC method for the inner current controller of non-inverting buck-boost converters. The proposed method could be applied for all converter parameters at all operating points, since the used linear prediction model is achieved without averaging and linearizing at a fixed operating point. Therefore, a single unified current controller can be formulated in contrast to conventional multi-mode methods, which divide the operation into buck, transition (buck-boost), and boost mode.

The proposed MPC enables a full utilization of the buck-boost converter topology both during transients and at steady-state. During transients, a superior dynamic control performance is achieved by actively utilizing all four switches for all operating ranges even when the input voltage is close to the output voltage. At steady-state, a small ripple is obtained with a minimum number of switch actions such that a high power efficiency can be maintained at minimum losses. Simulations are carried out to compare the control performance of the proposed MPC to existing control methods. The analysis verifies that the SMPC method achieves a good steady-state behavior and reduces the output voltage fluctuation at load changes by more than 50% compared to other methods.

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